

Supplementary Information

Exact stability certificates for neural temporal knowledge graph completion

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S1 Full proof and numerical realization

For a fixed query, write $a_c = (1 - \lambda)(p_w - p_c)$, $u_{bc} = \lambda(e_{bw} - e_{bc})/S$ and $v_b = m_b/S$. The original gap is $G_c = a_c + \sum_b u_{bc}$. For any proper subset D ,

$$(1 - \sum_D v_b)(g_D(w) - g_D(c)) = a_c(1 - \sum_D v_b) + \sum_{b \notin D} u_{bc} \quad (1)$$

$$= G_c - \sum_{b \in D} (u_{bc} + a_c v_b). \quad (2)$$

All retained windows have positive total mass, so the multiplier is positive. The winner remains ahead of competitor c if this expression is positive, or if it is zero and $w < c$. A violation is witnessed by a negative expression, or by zero when $c < w$.

Under a budget of at most k , deleting a negative contribution cannot reduce this expression, and deleting a zero contribution leaves it unchanged. Among proper subsets, an optimizer therefore consists of the largest $\min(k, B - 1)$ positive contributions. This proves exactness for each competitor. A query is stable if and only if it is stable against every competitor because the winning entity is selected by the same deterministic total ordering before and after editing. For $k \geq B$, the all-deleted case is evaluated directly using p . With $B = 0$, no nontrivial edit exists. With $B = 1$, the only proper subset is the empty set. Both cases are included by this construction.

The first violating prefix length, minimized over competitors and combined with the complete-deletion case, gives the minimum number of windows required to change the answer. If no deletion changes it, the implementation records $B + 1$ as a sentinel and reports radius B . The certificate surplus is the minimum signed numerator for the requested budget. When complete deletion violates the answer, a negative neural fallback gap is also included. Surplus is used as a diagnostic feature; it is not asserted to equal the smallest unscaled post-edit margin.

An unsupported candidate has $e_{bo} = 0$ for every window. Every such candidate has score $(1 - \lambda)p_o$ whenever memory remains, and score p_o after complete deletion. Since $\lambda < 1$, ordering the unsupported candidates by neural probability and then entity identifier gives one representative that dominates all others for every deletion. Checking all supported candidates and that representative therefore preserves the full-vocabulary guarantee.

Long histories can have retained mass far smaller than the original total. Direct subtraction of the deleted mass from the total can round that mass to zero. The implementation sorts signed contributions, then sums retained masses from the end of the ordered array. It compares the directly reconstructed winner and competitor scores to resolve floating-point ties. All certificate arithmetic uses float64. Property tests compare the sparse result with a full-vocabulary implementation and independent exhaustive subset enumeration; they include empty histories, tied scores, shared support and masses as small as 10^{-35} . The theorem is an exact-arithmetic statement, and the numerical implementation is checked against direct edited-score evaluation.

S2 Calibration inputs and operating policy

The four nested feature groups are listed below. For every logistic fit, inputs are standardized on the 2,000-query calibration-fit subset. A constant feature uses unit scale. Ridge regularization is chosen from $\{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1\}$ by negative log likelihood on the separate 1,000-query tuning subset. The belief feature group is selected by tuning AURC. The exact fitted means, scales, coefficients, penalty and operating thresholds are supplied in each frozen configuration file.

Probability calibration Hybrid maximum score.

Score calibration Neural maximum probability, neural probability of the hybrid winner, neural top-two margin, negative neural entropy, hybrid maximum score and hybrid top-two margin.

Evidence calibration The score group plus log event count, the winner’s fraction of evidence mass, temporal-window diversity $1 - \sum_b (m_b/S)^2$, neural-hybrid winner agreement, and the fraction of memory mass in the four latest occupied-window indices relative to the latest window.

Certificate calibration The evidence group plus one- and two-window surpluses, one- and two-window certificate indicators, and certificate radius divided by $\max(1, B)$.

The last recency feature uses the latest window and the three preceding bin indices, including empty intervening bins. It does not select the four most recent nonempty windows. Empty-history descriptors use zero memory fractions.

The three-seed ensemble is a selection control on the fixed hybrid winner of each evaluated seed. Inputs include its mean neural probability, probability variance, average-distribution entropy, vote fraction and mutual information. These five quantities augment the score feature group. Thus this comparator changes the confidence ranking, not the answer being judged. It uses three encoders, whereas the deployed certified-belief policy uses one.

Temperature scaling is fitted on the same calibration-fit role. Its score is the temperature-adjusted neural probability of the current hybrid winner. Brier score and ten equal-width-bin ECE refer to known-answer correctness of that winner. They are binary correctness calibration metrics, not a claim that the full categorical distribution is a calibrated probability over all unknown facts.

S3 Statistical definitions

For n queries ordered by decreasing confidence, with binary error e_i ,

$$\text{AURC} = \frac{1}{n} \sum_{j=1}^n \frac{1}{j} \sum_{i=1}^j e_i.$$

Risk at coverage c uses the first $\lfloor cn \rfloor$ predictions. Ties in confidence preserve source order. For certified selection, the list contains only queries passing the requested certificate; risk is unavailable if it is shorter than $\lfloor cn \rfloor$. The requested coverage denominator always remains all n queries. Deployment thresholds instead retain every query whose score reaches the frozen threshold, so ties may change realized coverage.

For random-window deletion, the binary discrimination endpoint is whether at least one of eight independently sampled deletions changes the answer. The random-flip rate itself is the mean of those eight responses. Shifted-window, wide-window and single-event outcomes are binary. AUROC uses negative margin or negative surplus so that larger values indicate greater vulnerability. The target for joint error is the union of an incorrect initial answer and a worst-case one-window answer change.

Unique observed query timestamps are sorted and divided into 20 contiguous blocks with approximately equal numbers of unique timestamps. Each bootstrap draw samples 20 blocks with

replacement. The same block multiplicities are applied to both methods and all three fitted seeds. Ranking metrics retain answer-count weights, while correctness metrics weight queries equally. AURC is recomputed on the repeated-query population, including its altered prefix denominators. Intervals are percentile intervals from 2,000 draws. An interval for a gated operating point uses only draws in which every seed has enough certified queries; its valid-replicate count is archived. Training variation is separately reported as the sample standard deviation over three seeds.

S4 Data provenance and implementation

The release includes `data/DATA_MANIFEST.md` for immutable public archive URLs, SHA-256 values and source license metadata. The TFLEX distribution provides point-entity queries with multiple known answers. Training entries are expanded into events without deduplication; the recorded event entries are unique in all three processed caches. The two primary test panels contain 5,057 and 5,052 designated held-out answers, respectively. No designated ICEWS panel target is observed in training at its query timestamp. GDELT is used only for maintenance scaling because its query-format target lists overlap with training.

The TeRDy optimizer retains the author implementation’s nonfinite-gradient sanitization. Each training log reports the number of corrected gradient elements per epoch; losses and exported prediction scores are finite. The learning rates are 0.02 for ICEWS14 and 0.008 for ICEWS05-15. The exact regularizers and their weights are specified in the pinned adapter and run records. The compact factorization fits reuse archived 30-epoch checkpoints whose hashes and original configurations are provided with the release. The revised query correctness and calibration protocol was applied to these fixed models. TeRDy was trained for this revision using the pinned author implementation, with validation-selected checkpoints and resource-adapted embedding ranks. Validation truth sets contain training and validation facts only. A development version had used the globally complete truth set for validation filtering; the final adapter removes test-only answers, and all TeRDy fits and validation decisions were recomputed with the unchanged parameter grids. The correction and superseded-output separation are recorded in the repository. Neither model is presented as a reproduction of a published leaderboard score. The primary revised results use renormalized memory. Historical additive and fixed-denominator pilots are excluded from every table in this manuscript.

S5 Complete selector results

Table S1: TeRDy on ICEWS14. Three-seed means. AURC is on its natural scale; Brier and ECE assess binary answer correctness. Risk is a percentage.

Selector	AURC	Brier	ECE	Risk 20%	Risk 40%
max_probability	0.190	0.189	0.120	7.200	13.567
candidate_probability	0.190	0.189	0.121	7.200	13.567
neural_margin	0.193	–	–	7.033	13.717
negative_entropy	0.196	–	–	8.133	14.617
hybrid_max	0.198	0.188	0.117	9.533	15.150
hybrid_margin	0.200	–	–	9.367	15.217
Probability calibration	0.198	0.171	0.026	9.533	15.150
Score calibration	0.189	0.167	0.035	7.167	13.367
Evidence calibration	0.194	0.166	0.025	8.567	13.350
Certificate calibration	0.195	0.166	0.024	9.100	13.233
Ensemble calibration	0.185	0.161	0.021	7.467	13.033
ensemble_probability	0.184	0.160	0.018	7.200	12.850
Temperature scaling	0.187	0.164	0.033	6.833	13.167
Certified 1-window	–	–	–	8.933	14.700
Certified 2-window	–	–	–	8.933	14.700

Table S2: TeRDy on ICEWS05-15. Three-seed means. AURC is on its natural scale; Brier and ECE assess binary answer correctness. Risk is a percentage.

Selector	AURC	Brier	ECE	Risk 20%	Risk 40%
max_probability	0.238	0.187	0.062	10.133	18.967
candidate_probability	0.237	0.187	0.070	10.133	18.967
neural_margin	0.242	–	–	10.467	19.217
negative_entropy	0.249	–	–	11.633	21.050
hybrid_max	0.238	0.188	0.070	10.133	18.667
hybrid_margin	0.243	–	–	10.333	19.367
Probability calibration	0.238	0.183	0.028	10.133	18.667
Score calibration	0.237	0.182	0.032	10.100	18.833
Evidence calibration	0.237	0.182	0.031	10.100	18.750
Certificate calibration	0.238	0.182	0.029	9.800	18.917
Ensemble calibration	0.228	0.175	0.030	9.033	17.583
ensemble_probability	0.228	0.182	0.085	9.000	17.467
Temperature scaling	0.237	0.185	0.053	10.000	19.033
Certified 1-window	–	–	–	10.067	18.683
Certified 2-window	–	–	–	10.067	18.683

Table S3: CTF on ICEWS14. Three-seed means. AURC is on its natural scale; Brier and ECE assess binary answer correctness. Risk is a percentage.

Selector	AURC	Brier	ECE	Risk 20%	Risk 40%
max_probability	0.608	0.291	0.253	59.133	61.700
candidate_probability	0.545	0.298	0.283	50.267	50.383
neural_margin	0.594	–	–	57.567	59.750
negative_entropy	0.634	–	–	62.833	64.650
hybrid_max	0.515	0.223	0.112	46.100	50.017
hybrid_margin	0.512	–	–	46.433	49.933
Probability calibration	0.515	0.206	0.031	46.100	50.017
Score calibration	0.501	0.201	0.025	41.067	49.417
Evidence calibration	0.464	0.186	0.030	30.167	46.767
Certificate calibration	0.462	0.185	0.027	30.900	46.433
Ensemble calibration	0.426	0.162	0.041	28.400	39.450
ensemble_probability	0.509	0.289	0.280	45.967	48.433
Temperature scaling	0.532	0.271	0.247	47.467	49.467
Certified 1-window	–	–	–	30.967	47.783
Certified 2-window	–	–	–	33.033	52.667

Table S4: CTF on ICEWS05-15. Three-seed means. AURC is on its natural scale; Brier and ECE assess binary answer correctness. Risk is a percentage.

Selector	AURC	Brier	ECE	Risk 20%	Risk 40%
max_probability	0.510	0.292	0.222	44.467	51.183
candidate_probability	0.458	0.279	0.240	39.033	44.983
neural_margin	0.507	–	–	43.767	50.817
negative_entropy	0.528	–	–	46.400	53.700
hybrid_max	0.449	0.240	0.132	30.700	46.667
hybrid_margin	0.453	–	–	33.133	46.567
Probability calibration	0.449	0.216	0.058	30.700	46.667
Score calibration	0.425	0.204	0.047	26.300	40.817
Evidence calibration	0.394	0.187	0.026	22.567	35.917
Certificate calibration	0.388	0.184	0.022	21.433	35.283
Ensemble calibration	0.386	0.180	0.025	21.433	33.467
ensemble_probability	0.412	0.228	0.175	30.233	38.483
Temperature scaling	0.457	0.267	0.221	39.267	45.167
Certified 1-window	–	–	–	21.433	36.567
Certified 2-window	–	–	–	21.667	40.717

S6 Frozen choices

Table S5: Validation-selected fusion and confidence models.

Backbone / data	Seed	T	λ	Belief	Best control
TeRDy / ICEWS14	0	0.5	0.1	Probability	Ensemble
TeRDy / ICEWS14	1	1	0.1	Evidence	Evidence
TeRDy / ICEWS14	2	1	0.1	Evidence	Evidence
TeRDy / ICEWS05-15	0	1	0.1	Probability	Temperature
TeRDy / ICEWS05-15	1	1	0.1	Score	Ensemble prob.
TeRDy / ICEWS05-15	2	1	0.25	Score	Ensemble prob.
CTF / ICEWS14	0	4	0.75	Certificate	Ensemble
CTF / ICEWS14	1	4	0.5	Certificate	Ensemble

Table S5: Validation-selected fusion and confidence models.

Backbone / data	Seed	T	λ	Belief	Best control
CTF / ICEWS14	2	4	0.75	Certificate	Ensemble
CTF / ICEWS05-15	0	2	0.75	Certificate	Ensemble
CTF / ICEWS05-15	1	4	0.75	Certificate	Ensemble
CTF / ICEWS05-15	2	4	0.5	Certificate	Ensemble

S7 Paired temporal-block intervals

Table S6: The complete prespecified paired comparison family.

Group	Comparison	Estimate	Lower	Upper	Valid
CTF/ICEWS05-15	mrr_gain	0.0505	0.0429	0.0576	2000
CTF/ICEWS05-15	accuracy_gain	0.0630	0.0530	0.0725	2000
CTF/ICEWS05-15	auroc_certificate_minus_best	0.0025	-0.0027	0.0082	2000
CTF/ICEWS05-15	certified_coverage	0.5514	0.5323	0.5721	2000
CTF/ICEWS05-15	risk20_certificate_minus_best	0.0000	-0.0147	0.0120	2000
CTF/ICEWS05-15	risk20_gate_minus_belief	0.0000	0.0000	0.0003	2000
CTF/ICEWS05-15	joint20_gate_minus_belief	0.0000	-0.0007	0.0000	2000
CTF/ICEWS05-15	risk40_certificate_minus_best	0.0182	0.0065	0.0283	2000
CTF/ICEWS05-15	risk40_gate_minus_belief	0.0128	0.0058	0.0221	2000
CTF/ICEWS05-15	joint40_gate_minus_belief	-0.0405	-0.0443	-0.0359	2000
CTF/ICEWS05-15	risk80_certificate_minus_best	0.0062	0.0033	0.0099	2000
CTF/ICEWS05-15	risk80_gate_minus_belief	—	—	—	0
CTF/ICEWS05-15	joint80_gate_minus_belief	—	—	—	0
CTF/ICEWS14	mrr_gain	0.0632	0.0558	0.0714	2000
CTF/ICEWS14	accuracy_gain	0.0700	0.0599	0.0820	2000
CTF/ICEWS14	auroc_certificate_minus_best	0.0357	0.0283	0.0426	2000
CTF/ICEWS14	certified_coverage	0.5916	0.5669	0.6199	2000
CTF/ICEWS14	risk20_certificate_minus_best	0.0250	0.0090	0.0408	2000
CTF/ICEWS14	risk20_gate_minus_belief	0.0007	-0.0013	0.0041	2000
CTF/ICEWS14	joint20_gate_minus_belief	-0.0033	-0.0057	-0.0006	2000
CTF/ICEWS14	risk40_certificate_minus_best	0.0698	0.0547	0.0817	2000
CTF/ICEWS14	risk40_gate_minus_belief	0.0135	-0.0022	0.0318	2000
CTF/ICEWS14	joint40_gate_minus_belief	-0.1107	-0.1289	-0.0936	2000
CTF/ICEWS14	risk80_certificate_minus_best	0.0085	0.0039	0.0138	2000
CTF/ICEWS14	risk80_gate_minus_belief	—	—	—	0
CTF/ICEWS14	joint80_gate_minus_belief	—	—	—	0
TeRDy/ICEWS05-15	mrr_gain	0.0044	0.0020	0.0069	2000
TeRDy/ICEWS05-15	accuracy_gain	0.0038	0.0006	0.0069	2000
TeRDy/ICEWS05-15	auroc_certificate_minus_best	0.0088	0.0066	0.0111	2000
TeRDy/ICEWS05-15	certified_coverage	0.8925	0.8839	0.9015	2000
TeRDy/ICEWS05-15	risk20_certificate_minus_best	0.0080	0.0000	0.0171	2000
TeRDy/ICEWS05-15	risk20_gate_minus_belief	0.0000	0.0000	0.0000	2000
TeRDy/ICEWS05-15	joint20_gate_minus_belief	0.0000	0.0000	0.0000	2000
TeRDy/ICEWS05-15	risk40_certificate_minus_best	0.0103	0.0032	0.0160	2000
TeRDy/ICEWS05-15	risk40_gate_minus_belief	-0.0002	-0.0004	0.0005	2000
TeRDy/ICEWS05-15	joint40_gate_minus_belief	-0.0003	-0.0010	0.0000	2000
TeRDy/ICEWS05-15	risk80_certificate_minus_best	0.0047	0.0013	0.0078	2000
TeRDy/ICEWS05-15	risk80_gate_minus_belief	0.0067	0.0047	0.0089	1748
TeRDy/ICEWS05-15	joint80_gate_minus_belief	-0.0096	-0.0115	-0.0076	1748
TeRDy/ICEWS14	mrr_gain	-0.0011	-0.0021	0.0000	2000
TeRDy/ICEWS14	accuracy_gain	0.0001	-0.0014	0.0018	2000
TeRDy/ICEWS14	auroc_certificate_minus_best	0.0082	0.0063	0.0101	2000

Table S6: The complete prespecified paired comparison family.

Group	Comparison	Estimate	Lower	Upper	Valid
TeRDy/ICEWS14	certified_coverage	0.9675	0.9625	0.9721	2000
TeRDy/ICEWS14	risk20_certificate_minus_best	0.0183	0.0134	0.0236	2000
TeRDy/ICEWS14	risk20_gate_minus_belief	0.0000	0.0000	0.0000	2000
TeRDy/ICEWS14	joint20_gate_minus_belief	0.0000	0.0000	0.0000	2000
TeRDy/ICEWS14	risk40_certificate_minus_best	0.0022	-0.0007	0.0068	2000
TeRDy/ICEWS14	risk40_gate_minus_belief	0.0000	0.0000	0.0000	2000
TeRDy/ICEWS14	joint40_gate_minus_belief	0.0000	0.0000	0.0000	2000
TeRDy/ICEWS14	risk80_certificate_minus_best	-0.0001	-0.0016	0.0017	2000
TeRDy/ICEWS14	risk80_gate_minus_belief	-0.0003	-0.0019	0.0016	2000
TeRDy/ICEWS14	joint80_gate_minus_belief	-0.0028	-0.0042	-0.0014	2000

Valid is the number of usable bootstrap draws out of 2,000. Gated operating points require sufficient certified queries in every seed. Estimates and intervals use natural metric units, not percentage points.

S8 Validation-threshold deployment

Thresholds are chosen on the operating-validation subset at nominal 20% and 40% coverage and then applied unchanged to the test panel. Certified selection adds the one-window gate to the same belief threshold. Values are percentages, mean $\pm$ sample SD over three fits.

Table S7: Test performance at frozen operating-validation thresholds.

Group	Policy	Nominal	Test coverage	Test error
TeRDy/ICEWS14	Belief	20	11.15 $\pm$ 2.06	5.16 $\pm$ 0.66
TeRDy/ICEWS14	Belief	40	34.45 $\pm$ 7.45	13.55 $\pm$ 1.42
TeRDy/ICEWS14	Certified	20	11.15 $\pm$ 2.06	5.16 $\pm$ 0.66
TeRDy/ICEWS14	Certified	40	34.45 $\pm$ 7.45	13.55 $\pm$ 1.42
TeRDy/ICEWS05-15	Belief	20	25.11 $\pm$ 0.63	11.82 $\pm$ 1.37
TeRDy/ICEWS05-15	Belief	40	50.29 $\pm$ 0.23	23.88 $\pm$ 0.32
TeRDy/ICEWS05-15	Certified	20	25.11 $\pm$ 0.63	11.82 $\pm$ 1.37
TeRDy/ICEWS05-15	Certified	40	50.09 $\pm$ 0.27	23.80 $\pm$ 0.27
CTF/ICEWS14	Belief	20	31.13 $\pm$ 1.43	39.91 $\pm$ 1.57
CTF/ICEWS14	Belief	40	56.61 $\pm$ 0.87	54.73 $\pm$ 0.85
CTF/ICEWS14	Certified	20	25.81 $\pm$ 1.05	37.66 $\pm$ 1.35
CTF/ICEWS14	Certified	40	31.05 $\pm$ 1.68	43.19 $\pm$ 2.34
CTF/ICEWS05-15	Belief	20	30.67 $\pm$ 1.79	29.25 $\pm$ 1.40
CTF/ICEWS05-15	Belief	40	51.71 $\pm$ 2.68	42.81 $\pm$ 2.12
CTF/ICEWS05-15	Certified	20	29.42 $\pm$ 0.69	28.59 $\pm$ 0.45
CTF/ICEWS05-15	Certified	40	39.55 $\pm$ 2.43	36.65 $\pm$ 1.14

S9 Intervention discrimination by fitted seed

AUROC is computed independently within each fitted seed, then summarized as mean $\pm$ sample SD. The paired intervals in the main figure instead pool query–seed predictions and resample timestamp blocks jointly. The two summaries describe different sources of variation.

Table S8: Within-seed intervention discrimination and between-seed variation.

Group	Intervention	Margin AUROC	Surplus AUROC
TeRDy/ICEWS14	random_flip2	0.926 ± 0.043	0.991 ± 0.007
TeRDy/ICEWS14	shifted_flip	0.926 ± 0.044	0.997 ± 0.002
TeRDy/ICEWS14	wide_flip	0.926 ± 0.043	0.998 ± 0.001
TeRDy/ICEWS14	event_flip	0.918 ± 0.050	0.994 ± 0.004
TeRDy/ICEWS05-15	random_flip2	0.853 ± 0.065	0.970 ± 0.013
TeRDy/ICEWS05-15	shifted_flip	0.884 ± 0.051	0.993 ± 0.003
TeRDy/ICEWS05-15	wide_flip	0.886 ± 0.051	0.994 ± 0.004
TeRDy/ICEWS05-15	event_flip	0.863 ± 0.058	0.987 ± 0.008
CTF/ICEWS14	random_flip2	0.352 ± 0.015	0.904 ± 0.004
CTF/ICEWS14	shifted_flip	0.394 ± 0.012	0.957 ± 0.001
CTF/ICEWS14	wide_flip	0.389 ± 0.011	0.969 ± 0.002
CTF/ICEWS14	event_flip	0.410 ± 0.006	0.957 ± 0.001
CTF/ICEWS05-15	random_flip2	0.464 ± 0.041	0.847 ± 0.018
CTF/ICEWS05-15	shifted_flip	0.599 ± 0.046	0.974 ± 0.003
CTF/ICEWS05-15	wide_flip	0.601 ± 0.051	0.982 ± 0.003
CTF/ICEWS05-15	event_flip	0.580 ± 0.039	0.957 ± 0.004

S10 Analytical controls and computation

Table S9: One-window controls on nonempty-memory test queries, seed 0. Exact and bound columns are certified percentages. Probe columns give the percentage of passed queries that fail the exact certificate, not the proportion of all queries.

Group	n	Exact	Bound	8 probes	32 probes	64 probes
TeRDy/ICEWS14	3556	98.82	96.37	0.06	0.03	0.00
TeRDy/ICEWS05-15	4335	92.48	80.39	2.65	1.23	0.67
CTF/ICEWS14	3556	40.66	38.39	21.07	3.98	1.57
CTF/ICEWS05-15	4335	49.37	46.87	31.93	18.60	11.35

Table S10: Certificate microbenchmark. Times are microseconds; speedup is the median paired dense/sparse ratio over three repetitions per query. The sparse implementation returns budgets, radius and witnesses; the dense control returns budget indicators. Empty-memory queries are retained in the All stratum.

Group	Memory	Queries	Sparse	Dense	Speedup
TeRDy/ICEWS14	All	32	164.05	1317.25	4.63
TeRDy/ICEWS14	Nonempty	22	473.90	2576.85	7.50
TeRDy/ICEWS05-15	All	32	335.25	3813.05	11.75
TeRDy/ICEWS05-15	Nonempty	25	486.40	8114.90	17.52
CTF/ICEWS14	All	32	177.70	1375.00	6.14
CTF/ICEWS14	Nonempty	22	274.80	2415.40	7.67
CTF/ICEWS05-15	All	32	246.35	3933.90	11.61
CTF/ICEWS05-15	Nonempty	25	444.50	11312.70	15.80

S10.1 Complete evidence-weight response

All prespecified weights are reported for seed 0, with temperature held at its validation-selected value. Values other than λ are percentages. Changed denotes disagreement with the neural-only answer.

Table S11: Prediction and stability over the evidence-weight grid.

Group	λ	MRR	Accuracy	Stable 1	Stable 2	Changed
TeRDy/ICEWS14	0	66.41	58.46	100.00	100.00	0.00
TeRDy/ICEWS14	0.1	66.28	58.54	99.16	98.92	0.92
TeRDy/ICEWS14	0.25	66.05	58.46	97.10	96.32	2.32
TeRDy/ICEWS14	0.5	64.25	55.50	82.70	76.40	11.74
TeRDy/ICEWS14	0.75	60.35	48.82	68.42	60.78	24.42
TeRDy/ICEWS05-15	0	63.64	54.62	100.00	100.00	0.00
TeRDy/ICEWS05-15	0.1	64.34	55.24	93.48	91.28	5.70
TeRDy/ICEWS05-15	0.25	63.56	53.90	81.14	76.36	16.36
TeRDy/ICEWS05-15	0.5	60.65	49.06	63.32	54.88	30.38
TeRDy/ICEWS05-15	0.75	57.34	44.78	55.74	45.24	39.64
CTF/ICEWS14	0	36.36	26.76	100.00	100.00	0.00
CTF/ICEWS14	0.1	42.55	32.52	74.00	68.02	28.36
CTF/ICEWS14	0.25	43.03	33.48	67.28	60.18	36.88
CTF/ICEWS14	0.5	43.01	33.86	60.98	52.96	43.78
CTF/ICEWS14	0.75	42.88	33.98	57.80	49.70	48.52
CTF/ICEWS05-15	0	45.49	33.00	100.00	100.00	0.00
CTF/ICEWS05-15	0.1	47.41	33.68	98.30	97.64	1.88
CTF/ICEWS05-15	0.25	48.37	34.98	92.28	89.90	7.30
CTF/ICEWS05-15	0.5	49.89	37.90	68.82	60.14	29.36
CTF/ICEWS05-15	0.75	50.34	39.18	56.10	45.44	46.22

S11 Complete maintenance measurements

The timing table contains 20 transactions per row, comprising ten deletion/restoration pairs. Affected is the median number of registered queries refreshed locally out of 5,000. Zero is the percentage of sampled transactions affecting no registered query. Times include event writes and refresh, and speedup is the median paired full/local ratio.

Table S12: Full and local refresh for every transaction size.

Dataset	Events	Full (ms)	Local (ms)	Speedup	Affected	Zero (%)
ICEWS14	1	1299.68	0.01	203862.64	0	80
ICEWS14	10	1346.99	2.49	502.57	7.5	0
ICEWS14	100	1381.96	48.99	25.91	120.5	0
ICEWS14	1000	1367.38	329.94	4.18	739	0
ICEWS05-15	1	2830.93	0.01	457948.11	0	60
ICEWS05-15	10	2983.28	30.39	107.72	20.5	0
ICEWS05-15	100	3260.87	612.99	5.41	145.5	0
ICEWS05-15	1000	4030.25	2406.99	1.67	973.5	0
GDELT	1	2071.07	0.49	4302.40	1	20
GDELT	10	2157.98	4.78	467.25	6	0
GDELT	100	2148.66	66.68	31.55	79	0
GDELT	1000	2160.59	503.70	4.29	631.5	0

Table S13: One-time cache construction and independent raw-event rebuild checks. Array memory is MiB for probabilities, outputs, object/window indices, masses and multiplicities; it excludes Python containers and the reverse index. RSS is the measured process-memory increment in MiB. The final column is the largest absolute difference across the 128 rebuilt queries; discrete outputs agree exactly.

Dataset	History entries	Build (s)	Arrays	RSS	Rebuild error
ICEWS14	72,826	2.20	283.35	434.45	2.2e-16
ICEWS05-15	368,962	5.02	700.48	950.59	2.2e-16
GDELT	2,308,165	5.46	275.41	304.77	5.6e-17

S11.1 Fixed-denominator comparison

The historical fixed-denominator certificate is evaluated on the same unedited score. Its one-window decision differs from the renormalized certificate for 0.14% (TeRDy, ICEWS14), 2.14% (TeRDy, ICEWS05-15), 0.82% (CTF, ICEWS14), 8.22% (CTF, ICEWS05-15) of all test queries. These decisions refer to different edited-score definitions; the renormalized definition is the one used for every primary result.

S12 Reproduction entry points

From a clean checkout, the commands in `README_NEUROCOMPUTING.md` reconstruct data, obtain the pinned author code, train or import the fixed backbones, freeze validation decisions, evaluate the test panels and regenerate the figures and PDFs. A separate build-only route starts from the archived query-level source data and needs no model training. No credentials are needed for the public archives. Full model training requires a CUDA-enabled PyTorch installation; certificate, statistics and figure reconstruction run on CPU.